

OMAR FADLALLAH

AI Engineer | LLM Systems | RAG Architectures | NLP

Mansoura, Egypt • +20 127 745 7703 • omayaser311@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Production-oriented AI Engineer with hands-on experience building and deploying LLM systems, hybrid RAG pipelines, and fine-tuned NLP models. Engineered an enterprise multi-tenant RAG platform achieving **sub-50ms P50 latency** and measurable cost reductions. Achieved a **70% CER reduction** on Arabic legal OCR through fine-tuning a 4B vision-language model. Interned at Devoteam delivering enterprise AI proof-of-concepts; **NVIDIA DLI-certified** in Applied Deep Learning and Generative AI.

TECHNICAL SKILLS

LLM & RAG: Retrieval-Augmented Generation, Prompt Engineering, LLM Fine-tuning (LoRA / QLoRA), Agentic Systems, Hybrid Retrieval, Cross-encoder Re-ranking, Multi-step Reasoning

Frameworks: LangChain, LangGraph, HuggingFace Transformers, PyTorch, FastAPI, Gradio

Vector & Search: Qdrant, FAISS, BM25, Semantic Search, Redis (semantic cache)

Evaluation & MLOps: MLflow, Prometheus, Grafana, Celery, ZenML, Docker, vLLM

Data & Storage: PostgreSQL, MySQL, MongoDB, Redis, Pandas, NumPy, SQLAlchemy

Languages: Python (primary), SQL, C++, R • Arabic (native), English (fluent)

Certifications: NVIDIA DLI – Applied Deep Learning • NVIDIA DLI – Generative AI • AWS Practical Training (DataCamp)

PROFESSIONAL EXPERIENCE

AI Engineer Intern • Devoteam

Dec 2025 – Feb 2026

- Fine-tuned BERT-based transformer models (PyTorch) for enterprise text classification and summarization, achieving a ~15% F1 improvement over baselines on enterprise evaluation sets of 10,000+ labeled samples.
- Built robust RAG ingestion pipelines, parsing and embedding 50,000+ enterprise document chunks to optimize vector search quality and retrieval accuracy.
- Developed and optimized prompt engineering techniques for 2 enterprise AI proof-of-concepts (PoCs), improving answer groundedness and response alignment against client-defined evaluation rubrics.

Generative AI Trainee • NVIDIA Deep Learning Institute & ITI

Aug 2025 – Oct 2025

- Developed 3 end-to-end Generative AI applications covering LLM deployment, RAG pipelines, and NLP fine-tuning over an intensive applied curriculum.
- Applied prompt engineering and context window optimization techniques, reducing hallucination rate by approximately 20% as measured on internal benchmark evaluation sets.
- Earned 2 NVIDIA Academy certifications: Applied Deep Learning and Generative AI with LLMs.

KEY PROJECTS

Computational Drug Discovery & Repurposing Platform ★ Graduation Project — Lead AI & Backend Engineer • Team of 9

Flask • FAISS-IVF • RDKit • Chemprop • PyTorch Lightning • DeepPurpose • OpenMM • AmberTools • PyTraj • Docker • asyncio • ThreadPoolExecutor

- Architected a production-grade, end-to-end computational biology platform unifying Chemical RAG, ADMET prediction, drug repurposing screening, and molecular dynamics simulation into a single async backend.
- Chemical RAG at Scale: Built FAISS-IVF vector index over 1M+ PubChem compounds with 2048-bit Morgan Fingerprints — sub-100ms similarity retrieval. Enriched Llama-3.1-8B context with RDKit-extracted physicochemical properties (MW, LogP, HBD/HBA, rotatable bonds, aromatic rings) instead of fine-tuning, giving the model reliable scientific grounding for similarity reasoning.
- ADMET Optimization (10s → 150ms on CPU): Evolved 5 MPNN models (Chemprop + PyTorch Lightning) from sequential pipeline (~10–12s) to fully async concurrent inference via asyncio — achieving 50–150ms single-molecule prediction and 1–2s batch processing on CPU only, no GPU dependency.
- Drug Repurposing Pipeline: High-throughput screening of 5,000+ approved drugs against protein targets using DeepPurpose binding-affinity models, with automated enrichment threads querying Open Targets and UniProt APIs.
- Molecular Dynamics: Productized a 9-step scientific protocol (PDBFixer → AmberTools → OpenMM → PyTraj RMSD/cross-correlation analysis) into a single autonomous API with ThreadPoolExecutor job queues and real-time status updates, deployed via multi-stage Docker with GitHub Actions CI/CD and integration testing.

Atlas AI Platform — Enterprise Multi-Tenant RAG — github.com/mirofadralla/Atlas-AI-Platform | [Demo](#) / [Python](#) · [LangChain](#) · [LangGraph](#) · [Qdrant](#) · [FAISS](#) · [Redis](#) · [FastAPI](#) · [Docker](#) · [Prometheus](#) · [Grafana](#)

- Designed a multi-tenant SaaS RAG platform with JWT authentication and strict per-tenant data isolation, enabling secure document Q&A for multiple organizations on a shared infrastructure.
- Built hybrid retrieval pipeline combining dense vector search (Qdrant), BM25 keyword matching, and cross-encoder re-ranking — improving answer relevance and reducing hallucinations compared to single-retriever baselines.
- Implemented a 3-tier caching architecture (in-memory → Redis semantic cache → database) achieving sub-50ms P50 API latency under load and substantially reducing LLM API call costs.
- Fine-tuned Qwen2.5-1.5B with LoRA/QLoRA on Arabic instruction-following data, improving model performance on domain-specific Arabic Q&A tasks.
- Added per-request cost and token tracking with real-time observability via Prometheus and Grafana (latency, token usage, drift alerts).
- Built a LangGraph-based agentic reasoning layer supporting multi-step query decomposition and tool use for complex questions.

Arabic Legal OCR — Fine-tuning Vision-Language Models for Structured Extraction — github.com/mirofadralla/ArabicOCR-VLM-Pipeline | [PyTorch](#) · [LlamaFactory](#) · [LoRA/QLoRA](#) · [vLLM](#) · [Gemma-3-4b-it](#) · [HuggingFace PEFT](#)

- Fine-tuned Gemma-3-4b-it (4B params) using LoRA on Arabic legal documents, reducing Character Error Rate from 0.74 to 0.22 — a 70% CER improvement — and lifting OCR accuracy from ~26% to ~78%.
- Scraped and processed 72+ legal documents from the Saudi Ministry of Justice API; used knowledge distillation with Gemini 1.5 Pro to generate structured JSON training labels.
- Built a dual-task extraction pipeline outputting structured document content and metadata (issuing authority, dates, signatures) with confidence scoring.
- Deployed production inference via vLLM with multi-LoRA support; trained with 4-bit NF4 quantization on a consumer RTX 4050 GPU.

LLMOps Platform — Production Prompt Management System — github.com/mirofadralla/LLM-Platform-LLMOps-System | [Demo](#) / [Python](#) · [FastAPI](#) · [React/Vite](#) · [Celery](#) · [PostgreSQL](#) · [Docker](#) · [Redis](#) · [LangChain](#)

- Built a full-stack LLMOps platform (FastAPI backend + React/Vite frontend) for managing, versioning, and evaluating LLM prompts — supporting API key authentication and Redis-backed rate limiting (60 req/min per key).
- Designed a modular evaluation engine with LLM-as-judge scoring, hallucination detection, and regression testing against golden examples for catching prompt regressions across versions.
- Implemented per-run cost and token tracking with prompt version diffing and rollback capability.
- Engineered async evaluation workflows with Celery + Redis; data layer uses PostgreSQL/SQLAlchemy with Alembic migrations.

EDUCATION

B.Sc. Computer Science · [Faculty of Computers and Information, Mansoura University](#) Oct 2022 – Jun 2026
GPA: 3.7 / 4.0 · Top 5% of cohort · Expected graduation: June 2026